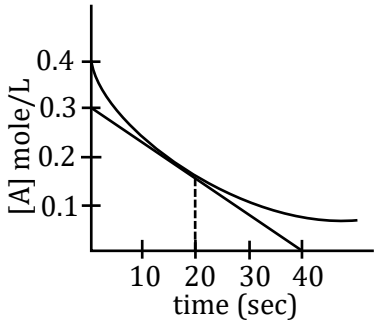
EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

Rate of reaction

1. For the reaction : 2A ⟶ product



Then rate of reaction is

(A) 7.5×10^{-3} M/sec. (B) 3.75×10^{-3} M/sec.

(C) 1.5×10^{-3} M/sec. (D) 2×10^{-3} M/sec.

CCK084

2. The rate of a reaction is expressed in different ways as follows :

$$+\frac{1}{2}\frac{d[C]}{dt} = -\frac{1}{3}\frac{d[D]}{dt} = +\frac{1}{4}\frac{d[A]}{dt} = -\frac{d[B]}{dt}$$

The reaction is:

(A) 4 A + B ⟶ 2C + 3D (B) B + 3 D ⟶ 4 A + 2 C

(C) A + B ⟶ C + D (D) B + D ⟶ A + C

CCK090

Order and Molecularity of Reactions

3. Which of the following rate law has an overall order of 0.5 for reaction involving substances x, y and z ?

(A) Rate = k $[C_x]$ $[C_y]$ $[C_z]$ (B) Rate = k $[C_x]^{0.5}[C_y]^{0.5}[C_z]^{0.5}$

(C) Rate = k $[C_x]^{1.5}$ $[C_y]^{-1}[C_z]^{\circ}$ (D) Rate = k $[C_x][C_z]^{\circ}$ / $[C_y]^2$

CCK091

4. The rate law of the reaction : A + 2B → product(P) is given by $\frac{d[P]}{dt}$ = K$[A]^2$ [B]. If A is taken in large excess, the order of the reaction will be –

(A) Zero (B) 1 (C) 2 (D) 3

CCK104

Zero Order reactions

5. K for a zero order reaction is 2×10^{-2} mol L^{-1} s^{-1}. If the concentration of the reactant after 25 s is 0.5 M, the initial concentration must have been.

(A) 0.5 M (B) 1.25 M (C) 12.5 M (D) 1.0 M

CCK087

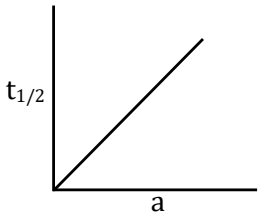
6. Half life ($t_{1/2}$) and completion time (T) of the zero order reaction are while k = 0.001 mol/litre / sec and a = 1M.

(A) 500 min, 750 min (B) 500 sec, 750 sec (C) 500 sec, 1000 sec (D) None of these

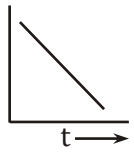
CCK089

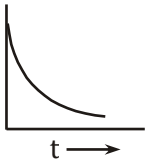
Rankers Academy JEE
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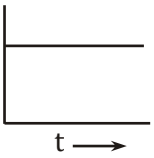
7. Consider the reaction A ⟶ B, graph between half life ($t_{1/2}$) and initial concentration (a) of the reactant is

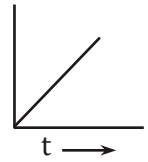


Hence graph between $-\frac{d[A]}{dt}$ and time will be

(A) 

(B) 

(C) 

(D) 

CCK088

8. H_2 gas is adsorbed on the metal surface like tungsten. This follows ..... order reaction -

(A) Third (B) Second (C) Zero (D) First

CCK092

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First order reactions

9. A reaction is of first order. After 100 minutes, 75 g of the reactant A are decomposed when 100g are taken initially. Calculate the time required when 150g of the reactant A are decomposed, the initial weight taken is 200g.

(A) 100 minutes (B) 200 minutes (C) 150 minutes (D) 175 minutes

CCK102

10. The rate constant of a first order reaction is $4 \times 10^{-3}\ s^{-1}$. At a reactant concentration of 0.02 M, the rate of reaction would be –

(A) 8×10^{-5} M s^{-1} (B) 4×10^{-3} M s^{-1} (C) 2×10^{-1} M s^{-1} (D) 4×10^{-1} M s^{-1}

CCK094

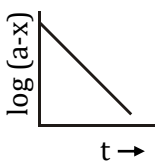
11. Correct statement about first order reaction is:-

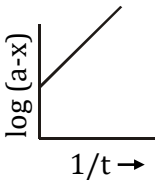
(A) $t_{completion}$ = finite (B) $t_{1/2} = \frac{a_0}{2k}$

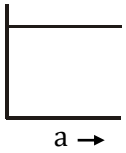
(C) Unit of k is mole lit^{-1} s^{-1} (D) $t_{1/2} \times k$ = constant (at given temperature)

CCK096

12. Which of the following curves represents a 1st order reaction :–

(A) 

(B) 

(C) t½ 

(D) A & C both

CCK095

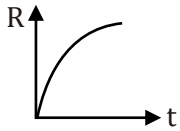
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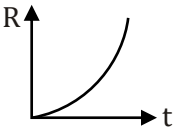
13. Given $t_{1/2}$ = 3 hr. Then how many gram of a substance will remain after 18 hr. from 300 gram of a substance -

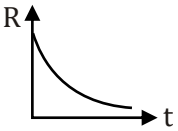
(A) 4.6 gm (B) 5.6 gm (C) 9.2 gm (D) 6.4 gm

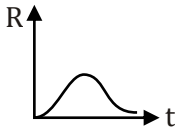
CCK097

14. If decomposition reaction A (g) ⟶ B (g) follows first order kinetics, then the graph of rate of formation (R) of B against time t will be

(A) 

(B) 

(C) 

(D) 

CCK103

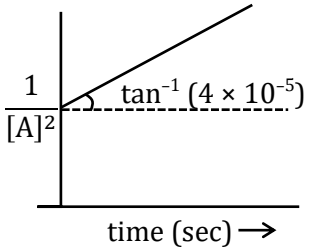
Second Order Reactions/nth order reaction

15. If 'a' is the initial concentration of the reactant, the half-life period of the reaction of nth order is inversely proportional to:

(A) a^{n-1} (B) a^{n} (C) a^{1-n} (D) a^{n+1}

CCK100

16. For a reaction A → Product



What is the value of k for the given reaction-

(A) $4 \times 10^{-5}\ M^{-1}\ s^{-1}$ (B) $\frac{4}{3} \times 10^{-5}\ M^{-2}\ s^{-1}$ (C) $2 \times 10^{-5}\ M^{-2}\ s^{-1}$ (D) $\frac{2}{3} \times 10^{-5}\ M^{-2}\ s^{-1}$

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CCK098

17. For a second order reaction, if the concentration of a reactant decreases from 0.08 M to 0.04 M in ten minutes, what would be the time taken for the concentration to decrease to 0.01 M :-

(A) 20 min. (B) 30 min. (C) 50 min. (D) 70 min.

CCK099

18. For a particular reaction with initial concentration of the reactants as a_1 and a_2, the half-life period are t_1 and t_2 respectively. The order of the reaction (n) is given by :-

(A) $n = 1 + \frac{\log(t_2/t_1)}{\log(a_2/a_1)}$ (B) $n = \frac{\log(t_1/t_2)}{\log(a_2/a_1)}$

(C) $n = 1 + \frac{\log(t_1/t_2)}{\log(a_2/a_1)}$ (D) None of these

CCK101

Calculation of First order rate constant using Pressure; Volume; Optical Rotation

19. For a first order homogeneous gaseous reaction, A ⟶ 2B + C, then initial pressure was P_i while total pressure after time 't' was P_t. The right expression for the rate constants k in terms of P_i, P_t and t is :

(A) $k = \frac{2.303}{t}\log\left(\frac{2P_i}{3P_i - P_t}\right)$ (B) $k = \frac{2.303}{t}\log\left(\frac{2P_i}{2P_t - P_i}\right)$

(C) $k = \frac{2.303}{t}\log\left(\frac{P_i}{P_i - P_t}\right)$ (D) none of these

CCK105

For More Join: @IITJEE_Advanced

20. The hydrolysis of sucrose was studied with the help of polarimeter and following data were

collected time (min.)	:	0	70	∞
observed rotation (degrees)	:	44	16.5	–11

The time taken when the reaction mixture will be optically inactive?
(Given : ln 2 = 0.7, ln 3 = 1.1, ln 5 = = 1.6)
(A) 16 min. (B) 69.47 min. (C) 160 min. (D) none of these

CCK107

21. At 100°C, the gaseous reaction, A → 3B + C is found to first order. Starting with pure A, if at the end of 20 minutes. the total pressure of the system is 190 mm and after a long time, when dissociation of A was complete, it was 400 mm. The partial pressure of A at the end of 20 min. is:-
(A) 100 mm (B) 30 mm (C) 70 mm (D) None of these

CCK109

22. In the following gaseous phase first order-reaction A(g) → 2B(g) + C(g) initial pressure was found to be 400 mm and it changed to 1000 mm after 20 minutes. Calculate rate constant in min^{-1}
(A) 0.1286 (B) 0.0693 (C) 0.693 (D) 1.286

CCK111

23. Azo isopropane decomposes according to the equation :–

$(CH_3)_2CHN{=}NCH(CH_3)_2(g) \xrightarrow{250\text{-}290^0C} N_2(g)+C_6H_{14}(g)$

It is found to be a first order reaction. If initial pressure is P_0 and pressure of the mixture at time t is (P_t) then rate constant K would be :–

(A) $k=\frac{2.303}{t}\log\frac{P_o}{2P_o-P_t}$ (B) $k=\frac{2.303}{t}\log\frac{P_o-P_t}{P_o}$

(C) $k=\frac{2.303}{t}\log\frac{P_o}{P_o-P_t}$ (D) $k=\frac{2.303}{t}\log\frac{2P_o}{2P_o-P_t}$

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CCK113

24. The decomposition of azo methane, at certain temperature according to the equation $(CH_3)_2N_2 \longrightarrow C_2H_6 + N_2$ is a first order reaction.
After 40 minutes from the start, the total pressure developed is found to be 350 mm Hg in place of initial pressure 200 mm Hg of azo methane. The value of rate constant k is :
(A) 2.88×10^{-4} sec^{-1} (B) 1.25×10^{-4} sec^{-1} (C) 5.77×10^{-4} sec^{-1} (D) none of these

CCK106

25. A ⟶ B first order reaction A is optical active and B is optically inactive. A series of experiment were conducted on a solution of A

Time (min.)	:	0	60	∞
Optical rotation :		82°	77°	2°

Assume some impurity is present calculate the optical rotation after 5 hours.
(Given ln 1.066 = 0.064, $e^{0.16}$ = 1.17)
(A) 60 (B) 30 (C) 20 (D) 120

CCK108

26. At a certain temperature, the half-life periods for the catalytic decomposition of NH_3 were found to be as follows :-

Pressure (mm Hg)	50	100	200
Half-life period (hrs.)	3.52	1.76	0.88

What will be the pressure when the half-life period is 2.5 hours.
(A) 117 (B) 206 (C) 70 (D) 160

CCK110

For More Join: @IITJEE_Advanced

27. At certain temperature, the half-life period for the thermal decomposition of a gaseous substance depends on the initial partial pressure of the substance as follows

P(mmHg)	500	250
$t_{1/2}$(in min.)	235	950

Find the order of reaction [Given log (23.5) = 1.37 ; log (95) = 1.97; log 2 = 0.30]

(A) 1 (B) 2 (C) 2.5 (D) 3

CCK112

28. For a 1st order homogeneous gaseous reaction

$A \rightarrow 2B + C$,

if the pressure after time t was P_t and after long time was P_∞, then rate constant (k) in terms of P_t & P_∞ and t is -

(A) $k = \frac{2.303}{t}\log\left(\frac{P_\infty}{P_\infty - P_t}\right)$ (B) $k = \frac{2.303}{t}\log\left(\frac{2P_\infty}{P_\infty - P_t}\right)$

(C) $k = \frac{2.303}{t}\log\left(\frac{2P_\infty}{3(P_\infty - P_t)}\right)$ (D) None of these

CCK114

Experimental determination of order of reaction

29. The experimental data for the reaction $2A + B_2 \rightarrow 2AB$ is

Exp.	$[A]_0$	$[B_2]_0$	Rate (mole s^{-1})
(i)	0.50	0.50	1.6×10^{-4}
(ii)	0.50	1.00	3.2×10^{-4}
(iii)	1.00	1.00	3.2×10^{-4}

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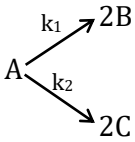
The rate equation for the above data is

(A) Rate = $k[B_2]$ (B) Rate = $k[B_2]^2$ (C) Rate = $k[A]^2[B_2]^2$ (D) Rate = $k[A]^2[B_2]$

CCK086

Parallel Reaction, Reversible Reaction, Sequential Reaction

30. For given hypothetical elementary parallel reaction,



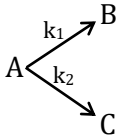
Where $\frac{k_1}{k_2} = \frac{1}{2}$

Initially only 2 moles of A are present. The total no. of moles of A, B and C at the end of 75% reaction are

(A) 2 (B) 3 (C) 4 (D) 3.5

CCK115

31. A Substance undergoes first order decomposition. The decomposition follows two parallel first order reactions as



$k_1 = 1.26 \times 10^{-4}\ s^{-1}$

$k_2 = 3.8 \times 10^{-5}\ s^{-1}$

The percentage distribution of B and C are

(A) 80% B and 20% C (B) 60% B and 40% C

(C) 90% B and 10% C (D) 76.83% B and 23.17% C

CCK116

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Reversible Reaction

32. The rate constant for the forward reaction $A(g) \rightleftharpoons 2B(g)$ is $1.5 \times 10^{-3}\ s^{-1}$ at 100 K. If 10^{-5} moles of A and 100 moles of B are present in a 10 litres vessel at equilibrium, then rate constant for the backward reaction at this temperature is

(A) $1.50 \times 10^{4}\ L\ mol^{-1}\ s^{-1}$ (B) $1.5 \times 10^{11}\ L\ mol^{-1}\ s^{-1}$

(C) $1.5 \times 10^{10}\ L\ mol^{-1}\ s^{-1}$ (D) $1.5 \times 10^{-11}\ L\ mol^{-1}\ s^{-1}$

CCK117

Mechanism of Reaction

33. The reaction mechanism for the reaction $P \rightarrow R$ is as follows :

$P \underset{k_2}{\overset{k_1}{\rightleftharpoons}} 2Q$ (fast) ; $2Q + P \xrightarrow{k_3} R$ (slow)

the rate law for the main reaction ($P \rightarrow R$) is :

(A) $k_1 [P] [Q]$ (B) $k_1 k_2 [P]$ (C) $\frac{k_1 k_3 [P]^2}{k_2}$ (D) $k_1 k_2 [a]$

CCK121

34. The reaction of formation of phosgene from CO and Cl_2 is $CO + Cl_2 \rightarrow COCl_2$. The proposed mechanism is :-

(i) $Cl_2 \underset{k_2}{\overset{k_1}{\rightleftharpoons}} 2Cl$ (ii) $Cl + CO \underset{k_4}{\overset{k_3}{\rightleftharpoons}} COCl$

(iii) $COCl + Cl_2 \xrightarrow{k_5} COCl_2 + Cl$ (slow)

Find the correct expression of rate law :-

(A) $r = k_5 \times \frac{k_3}{k_4} \times \left(\frac{k_1}{k_2}\right)^{1/2} [CO]\ [Cl_2]^{3/2}$ (B) $r = k_5 \times \frac{k_3}{k_4} \times \left(\frac{k_1}{k_2}\right)^{1/2} [CO]\ [Cl_2]^{1/2}$

(C) $r = k_5 \times \frac{k_3}{k_4} \times \left(\frac{k_1}{k_2}\right)^{1/2} [CO]\ [Cl_2]^{3/2}$ (D) None of these

CCK123

35. The reaction of ozone with oxygen atoms in the presence of chlorine atoms can occur by a two step process shown below :

$O_3(g) + Cl^{\bullet}(g) \rightarrow O_2(g) + ClO^{\bullet}(g)$ $k_i = 5.2 \times 10^{9}\ L\ mol^{-1} s^{-1}$ ...(i)

$ClO^{\bullet}(g) + O\cdot(g) \rightarrow O_2(g) + Cl^{\bullet}(g)$ $k_{ii} = 2.6 \times 10^{10}\ L\ mol^{-1} s^{-1}$ ...(ii)

The closest rate constant for the overall reaction $O_3(g) + O^{\bullet}(g) \rightarrow 2O_2(g)$ is :

(A) $3.1 \times 10^{10}\ L\ mol^{-1} s^{-1}$ (B) $2.6 \times 10^{10}\ L\ mol^{-1} s^{-1}$

(C) $5.2 \times 10^{9}\ L\ mol^{-1} s^{-1}$ (D) $1.4 \times 10^{20}\ L\ mol^{-1} s^{-1}$

CCK125

36. Following mechanism has been proposed for a reaction : $2A + B \rightarrow D + E$

Step - I : $A + B \rightarrow C + D$ (slow)

Step - II : $A + C \rightarrow E$ (fast)

The rate law expression for the reaction is –

(A) $r = k[A]^2[B]$ (B) $r = k[A]\ [B]$ (C) $r = k[A]^2$ (D) $r = k[A][C]$

CCK120

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37. The chemical reaction
$2O_3 \longrightarrow 3O_2$, proceeds as follows :
$O_3 \rightleftharpoons O_2 + O$ ...(fast)
$O + O_3 \rightarrow 2O_2$ ...(slow)
The overall order of the reaction is :-
(A) 1 (B) –1 (C) 0 (D) 2

CCK122

38. The mechanism of the reaction :
$2NO + O_2 \rightarrow 2NO_2$is
Step-I : $NO + NO \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} N_2O_2$
Step-II $N_2O_2 + O_2 \xrightarrow{k_2} 2NO_2$(slow)
the rate constant of the reaction is :
(A) k_2 (B) $k_2k_1(k_{-1})$ (C) k_2k_1 (D) $k_2\left(\frac{k_1}{k_{-1}}\right)$

CCK124

Temperature dependence of rate

39. If for two reaction $E_{a_1} > E_{a_2}$ & μ_1 & μ_2 are temperature coefficient respectively, then which of the following is correct :
(A) $\mu_1 > \mu_2$ (B) $\mu_1 < \mu_2$ (C) $\mu_1 = \mu_2$ (D) None of these

CCK131

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40. The rate of reaction is tripled for 10° rise in temperature. The increase in the reaction rate as a result of temperature rise from 20° to 60° is :-
(A) 81 times (B) 243 times (C) 729 times (D) 614 times

CCK132

41. The rate constant of a first order reaction at 37°C is 10^{-3} min^{-1}. The temperature coefficient of this reaction is 2. What is the rate constant (in min^{-1}) at 17°C for this reaction ?
(A) 10^{-3} (B) 5×10^{-4} (C) 2.5×10^{-3} (D) 2.5×10^{-4}

CCK130

Arrhenius equation

42. A large increase in the rate of a reaction for a rise in temperature is due to –
(A) Increase in the number of collisions (B) Increase in the number of activated molecules
(C) Lowering of activation energy (D) Shortening of the mean free path

CCK127

43. The plot of ln k versus 1/T is linear with slope of :
(A) $-E_a/R$ (B) E_a/R (C) $E_a/2.303\ R$ (D) $-E_a/2.303\ R$

CCK129

44. A first order reaction is 50% completed in 20 minutes at 27°C and in 5 min at 47°C. The energy of activation of the reaction is
(A) 100 kJ/mol (B) 55.14 kJ/mol (C) 11.97 kJ/mol (D) 6.65 kJ/mol

CCK134

For More Join: @IITJEE_Advanced

45. For the first order reaction A $\longrightarrow$ B + C, carried out at 27 °C if 3.8×10^{-16} % of the reactant molecules can overcome energy barrier, the E_a (activation energy) of the reaction is
[log 3.8 = 0.58, 2.303 × 8.314 × 17.42 = 333.33]
(A) 12 kJ/mole (B) 831.4 kJ/mole (C) 100 kJ/mole (D) 111.11 J/mole

CCK138

46. Given that K is the rate constant for any order reaction at temperature T, then the value of $\lim_{T\to\infty} \log k$.

(A) $\frac{A}{2.303}$ (B) A (C) 2.303 A (D) log A

CCK133

Collision Theory

47. For producing the effective collisions, the colliding molecules must possess:-
(A) A certain minimum amount of energy
(B) Energy equal to or greater than threshold energy
(C) Proper orientation
(D) Threshold energy as well as proper orientation of collision

CCK136

48. According to collision theory of reaction –
(A) Every collision between reactant leads to chemical reaction
(B) Rate of reaction is proportional to velocity of molecules
(C) All reactions which occur in gaseous phase are zero order reaction
(D) Rate of reaction is directly proportional to collision frequency.

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CCK142

Effect of Catalyst

49. On introducing a catalyst at 500K, the rate constant of a first order reaction increases 2.718 times. If the activation energy in the presence of a catalyst is 4.15 kJ mol^{-1}. Then what will be E_a in absence of catalyst (value of e = 2.718) :-
(A) 4.15 kJ (B) 2.08 kJ (C) 2.718 kJ (D) 8.3 kJ

CCK140

50. **Statement-1** : A catalyst provides an alternative path to the reaction in which conversion of reactants into products takes place quickly
Statement-2 : The catalyst forms an activated complex of lower potential energy, with the reactants by which more number of molecules are able to cross the barrier per unit of time.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

CCK144

For More Join: @IITJEE_Advanced

EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

Rate of reaction

1. For the reaction A + 3B → C, select the correct statement (s)-

(A) $\frac{d[C]}{dt} = \frac{-d[A]}{dt}$

(B) $\frac{3d[C]}{dt} = -\frac{d[B]}{dt}$

(C) Rate law must be r = k [A][B]3

(D) Units for rate of reaction are independent of order of reaction

CCK168

Zero Order reactions

2. Which of the following is/are **correct** statement?

(A) Stoichiometry of a reaction tells about the order of the elementary reactions.

(B) For a zero-order reaction, rate and the rate constant are identical.

(C) A zero-order reaction is controlled by factors other than concentration of reactants.

(D) A zero-order reaction is always an elementary reaction.

CCK169

First Order reactions

3. Thermal decomposition of compound X is a first order reaction. If 75% of X is decomposed in 100 min. How long will it take for 90% of the compound to decompose?

Given : log 2 = 0.30

(A) 190 min (B) 176.66 min (C) 166.66 min (D) 156.66 min

CCK170

Rankers Academy JEE

4. 2X (g) + Y(g) + 3Z (g) → Products. The rate equation of above reaction is given by :

Rate = K $[X]^1$ $[Y]^0$ $[Z]^2$.

Choose the correct statements

(A) If [z] >> [x] and 75% of X undergoes reaction in 20 sec, then 50% of X will react in 10 sec.

(B) Rate of reaction decreases by reducing the concentration of Y to half of the original value

(C) The half-life of Z increases by increasing its concentration if [x] >> [z]

(D) On doubling the concentration of each of X, Y & Z, rate of reaction becomes 8 times

CCK171

Calculation of First order rate constant using Pressure; Volume; Optical Rotation

5. For a first order reaction : A(g) → 2B(g)

Time (in second)	0	20	40	∞
Total pressure of system (in mm. of Hg)	64	112	124	128

(A) Half life of reaction is 10 sec

(B) Value of rate constant for reaction is 6.93×10^{-3} sec^{-1}

(C) Total pressure at t = 50 sec will be 126 mm of Hg

(D) Reaction must be a complex reaction

CCK172

6. Consider a reaction A(g) ⟶ 3B(g) + 2C(g) with rate constant 1.386×10^{-2} min^{-1}. Starting with 2 moles of A in 12.5 litre a closed vessel initially, if reaction is allowed to takes place at constant pressure & at 298K then find the concentration of B after 100 min.

(A) 0.04 M (B) 0.36 M (C) 0.09 M (D) None of these

CCK173

For More Join: @IITJEE_Advanced

Effect of catalyst

7. Which of the following is **incorrect** for first order reaction ?

(A) On introducing catalyst, both rate constant and rate of reaction increases.

(B) On increasing temperature both rate constant & rate of reaction increases.

(C) On decreasing volume both rate constant & rate of gaseous reaction increases.

(D) On increasing concentration of gaseous reactant at constant volume & constant temperature both total pressure and rate of the reaction increases.

CCK174

MATRIX MATCH TYPE QUESTION

Second order kinetics; n^{th} order kinetics

8. **Match the column :**

Column-I	**Column-II**
(A) Inversion of cane sugar in excess water.	(P) not 100% complete
(B) saponification reaction with 1M NaOH	(Q) pseudo-first order
(C) decomposition of HI on gold	(R) zero order
(D) radioactive decay	(S) second order

(A) A → P,Q ; B → S,P ; C → R ; D → P

(B) A → P ; B → P ; C → R ; D → P

(C) A → S,P ; B → P,Q ; C → P ; D → R

(D) A → P,R ; B → S,P ; C → P ; D → R

CCK175

9. For the reaction A + B → product, Given : $[A]_0 = [B]_0$

List-I (Observed Rate Law) is -

(P) $r = k[A]$

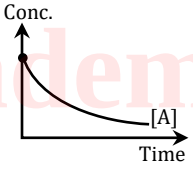
(Q) $r = k[A]^{1/2}[B]^{1/2}$

(R) $r = k[A]\,[B]$

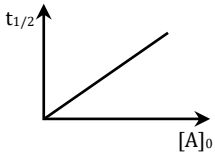
(S) $r = k[A]_0\,[B]_0$

List-II (Graph)

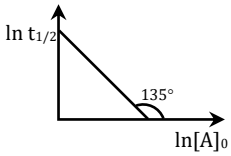
(1)



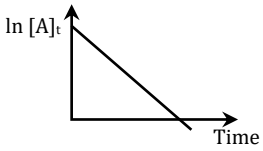
(2)



(3)



(4)



Code :

	P	**Q**	**R**	**S**
(A)	4	1	3	2
(B)	2	3	1	4
(C)	1	2	3	4
(D)	4	3	2	1

CCK176

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For More Join: @IITJEE_Advanced

EXERCISE (S)

Part-1 : JEE Main (Numerical Value type)

- The answer to each question is a **Numerical Value**.
- For each question, enter the correct integer value (In case of non-integer value, the answer should be **rounded off** to the nearest Integer).

Rate of reaction

1. The reaction $2A + B + C \rightarrow D + E$ is found to be first order in A, second order in B and zero order in C. How many times rate of reaction increase if concentrations of A, B, and C becomes double ?

CCK093

2. Ammonia and oxygen react at higher temperatures as

$4NH_3(g) + 5O_2(g) \longrightarrow 4NO(g) + 6H_2O(g)$

In an experiment, the concentration of NO increases by 1.08×10^{-2} mol litre^{-1} in 3 seconds. Calculate.

(i) If rate of reaction is $A \times 10^{-4}$ mol/litre then A will be.

(ii) If rate of disappearance of ammonia $B \times 10^{-4}$ mol/litre then B will be.

(iii) If rate of formation of water $B \times 10^{-4}$ mol/litre then B will be.

CCK149

3. At 27°C, for the reaction, $N_2O_{4_{(g)}} \rightarrow 2NO_{2_{(g)}}$, rate of reaction is 6×10^{-3} atm min^{-1}. If rate of same reaction in terms of mol L^{-1} min^{-1} is $y \times 10^{-4}$ mol L^{-1} min^{-1}, then calculate value of 5y

Rankers Academy JEE

CCK024

4. $xA + yB \rightarrow zC$. If $-\frac{d[A]}{dt} = -\frac{d[B]}{dt} = 1.5\frac{d[C]}{dt}$, then (x + y + z) is.

(x, y & z are the lowest integral value.)

CCK150

Zero Order reactions

5. The pressure of a gas decomposing at the surface of a solid catalyst has been measured at different time and the results are given below

t (sec)	0	100	200	300
Pr. (Pascal)	4×10^3	3.5×10^3	3×10^3	2.5×10^3

Determine the order of reaction and its rate constant in Pa/sec.

CCK159

6. $A(g) \rightarrow 2B(g)$ initially 2 moles of A are taken in 5 litre vessel. After 20 min, $[A]_t = \frac{[B]_t}{2}$. Find half life time of A in the first order reaction

CCK026

7. A drop of solution (volume 0.10 ml) contains 6×10^{-6} mole of $H^{\oplus}$. If the rate constant of disappearance of $H^{\oplus}$ is 1×10^7 mole litre^{-1} sec^{-1}, If time for complete disappearance for $H^{\oplus}$ is $x \times 10^{-10}$ sec, then x will be

CCK028

For More Join: @IITJEE_Advanced

8. For a zero order chemical reaction,

$2NH_3(g) \longrightarrow N_2(g) + 3H_2(g)$

rate of reaction = 0.1 atm/sec. Initially only NH_3 is present & its pressure = 3 atm. Calculate total pressure in atm at t = 10 sec.

CCK030

First order reactions

9. A first order reaction is 75% completed in 72 min. How long time in minute will it take for

(i) 50% completion (ii) 87.5% completion

CCK153

10. Calculate ratio of the time required for 99.9% of the reaction to take place to time required for 50.00% of the reaction to take place if reaction follows first order kinetics.

[Give your answer as nearest integer]

CCK154

11. Calculate the half-life in min of the first-order reaction

$C_2H_4O(g) \longrightarrow CH_4(g) + CO(g)$

if the initial pressure of $C_2H_4O(g)$ is 80 mm of Hg and the total pressure at the end of 20 minutes is 120 mm of Hg

CCK155

12. 10 gram mixture of two gases A_2 (mol. wt. = 20) and B_2 (mol. wt. = 30), which decompose by first order kinetics, was taken in a vessel. The half life of decomposition of A_2 and B_2 are 2 and 3 hours respectively. After 6 hours weight of mixture of A_2 and B_2 is found to be 2 gram. Find the weight of A_2 in the initial mixture-

Rankers Academy JEE

CCK022

Second order kinetics; nth order kinetics

13. If $t_{1/2}$ of a second order reaction is 1.0 hr. After what time in hr, the amount of reactant will be 25% of the initial amount?

CCK157

Calculation of rate constant using different parameters

14. For the 1st order reaction

$A \longrightarrow B + C$

Time	t	∞
Total pressure of (A + B + C)	P_2	P_3

If $k = \frac{l}{t}\ln\frac{P_3}{(xP_3 - yP_2)}$ then x + y will be.

CCK160

15. The following data were obtained in experiment of inversion of cane sugar.

Time (minutes)	0	60	120	180	∞
Angle of rotation (degree)	+15	+ 7	+ 3	+1	–1

After what time (in min) would you expect a zero reading in polarimeter?

CCK162

For More Join: @IITJEE_Advanced

Mechanism of Reaction

16. The reaction $2NO + Br_2 \longrightarrow 2NOBr$, is supposed to follow the following mechanism

(i) $NO + Br_2 \overset{fast}{\rightleftharpoons} NOBr_2$ (ii) $NOBr_2 + NO \xrightarrow{slow} 2NOBr$

total order of reaction will be

CCK166

17. For the reaction $2H_2 + 2NO \longrightarrow N_2 + 2H_2O$, the following mechanism has been suggested :

$2NO \rightleftharpoons N_2O_2$ equilibrium constant = K_1 (fast)

$N_2O_2 + H_2 \xrightarrow{k_2} N_2O + H_2O$ (slow)

$NO + H_2 \xrightarrow{k_3} N_2 + H_2O$ (fast)

Order of reaction w.r.t. NO will be

CCK167

Temperature dependence of rate

18. In gaseous reactions important for understanding the upper atmosphere, H_2O and O react bimolecularly to form two OH radicals. ΔH for this reaction is 72 kJ at 500 K and $E_a = 77$ kJ mol^{-1}, then calculate E_a in kJ/mole for the bimolecular recombination of 2OH radicals to form H_2O & O at 500 K

CCK164

Part-2 : JEE Advanced (Decimal type)

- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.

Rankers Academy JEE

Zero Order reactions

19. In the reaction (A $\longrightarrow$ B) rate constant is 1.2×10^{-2} M s^{-1}. What is concentration of B (in mole/litre) after 10 and 20 min., if we start with 10 M of A.

CCK151

20. From the following data for the zero order reaction : A $\longrightarrow$ products. Calculate the value of k in M min^{-1}.

Time (min.)	**[A]**
0.0	0.10 M
1.0	0.09 M
2.0	0.08 M

CCK152

SECOND ORDER REACTIONS/n^{th} order reaction

21. In the II order reaction : $2A \rightarrow A_2$. The rate of formation of A_2 is 10^{-5} M sec^{-1} at 0.01M concentration of A. Calculate the rate constant in the rate of disappearance of 'A' in M^{-1} sec^{-1}.

CCK156

For More Join: @IITJEE_Advanced

Calculation of rate constant using different parameters

22. The thermal decomposition of dimethyl ether was measured by finding the increase in pressure at 500°C in the reaction

$$(CH_3)_2O(g) \longrightarrow CH_4(g) + H_2(g) + CO(g)$$

Time (sec.)	200	400	∞
Change in Pressure (mm Hg)	540	594	600

The initial pressure of ether was 300 mm Hg. Determine the rate constant in $M^{1-n}sec^{-1}$ here is n is order of reaction.

CCK161

23. The decomposition of N_2O_5 according to the equation $2\,N_2O_5(g) \longrightarrow 4NO_2(g) + O_2(g)$ is a first order reaction. After 60 min. from start of decomposition in a closed rigid vessel, the total pressure developed is found to be 350 mm Hg. On complete decomposition, the total pressure is 500 mm Hg. Calculate the rate constant for disappearance of $N_2O_5(k_{N205})$ in min^{-1} for the reaction.

CCK163

Experimental Determination of Order of reaction

24. At 800° C, the rate of reaction, $2NO(g) + H_2(g) \longrightarrow N_2(g) + H_2O(g)$ changes with the concentration of NO(g) and $H_2(g)$ as-

Exp.no.	[NO] in M	$[H_2]$ in M	$-\frac{1}{2}\frac{d[NO]}{dt}$ in M sec^{-1}
(i)	1.5×10^{-3}	4×10^{-3}	4.50×10^{-9}
(ii)	1.5×10^{-3}	2×10^{-3}	2.25×10^{-9}
(iii)	3.0×10^{-3}	2×10^{-3}	9.00×10^{-9}

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(a) What is the order of reaction?

(b) What is the rate of reaction in M sec^{-1} when $[H_2]$ = 1.5 M and [NO] = 1.0M ?

CCK158

Temperature dependence of rate

25. The specific reaction rate for a reaction increases by a factor of 4, if the temperature is changed from 27°C to 47°C. Find the activation energy (in kcal) for the reaction (Use ln 2 = 0.7)

CCK165

For More Join: @IITJEE_Advanced

EXERCISE - JEE (Main) PYQ

1. Consider the elementary reaction **[JEE (Main) 2025]**

$A(g) + B(g) \rightarrow C(g) + D(g)$

If the volume of reaction mixture is suddenly reduced to $\frac{1}{3}$ of its initial volume, the reaction rate will become 'x' times of the original reaction rate. The value of x is :

(1) $\frac{1}{9}$ (2) 9 (3) $\frac{1}{3}$ (4) 3

CCK202

2. NO_2 required for a reaction is produced by decomposition of N_2O_5 in CCl_4 as by equation

$2N_2O_{5(g)} \rightarrow 4NO_{2(g)} + O_{2(g)}$

The initial concentration of N_2O_5 is 3 mol L^{-1} and it is 2.75 mol L^{-1} after 30 minutes.

The rate of formation of NO_2 is x × 10^{-3} mol L^{-1} min^{-1}, value of x is _____. **[JEE (Main) 2024]**

CCK177

3. At 518° C, the rate of decomposition of a sample of gaseous acetaldehyde, initially at a pressure of 363 Torr, was 1.00 Torr s^{-1} when 5% had reacted and 0.5 Torr s^{-1} when 33% had reacted. The order of the reaction is : **[JEE (Main) 2018]**

(1) 3 (2) 1 (3) 0 (4) 2

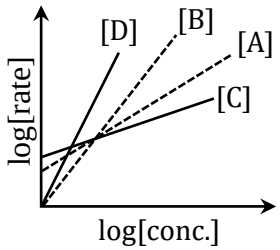
Rankers Academy JEE

CCK037

4. Consider the following reactions :

$A \rightarrow P1$; $B \rightarrow P2$; $C \rightarrow P3$; $D \rightarrow P4$

The order of the above reactions are a, b, c, and d, respectively. The following graph is obtained when log [rate] vs. log[conc.] are plotted :



Among the following, the correct sequence for the order of the reactions is: **[JEE (Main) 2020]**

(1) a > b > c > d (2) c > a > b > d (3) d > b > a > c (4) d > a > b > c

CCK045

5. Decomposition of X exhibits a rate constant of 0.05 µg/year. How many years are required for the decomposition of 5 µg of X into 2.5 µg ? **[JEE (Main) 2019]**

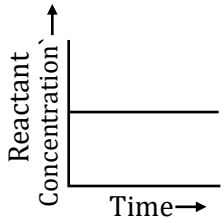
(1) 50 (2) 25 (3) 20 (4) 40

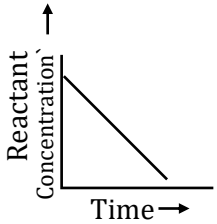
CCK042

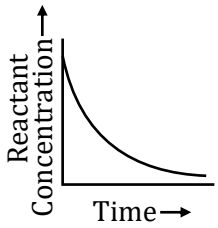
For More Join: @IITJEE_Advanced

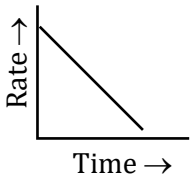
6. Which of the following graphs most appropriately represents a zero order reaction ?

[JEE (Main) 2025]

(1) 

(2) 

(3) 

(4) 

CCK203

7. The reaction $2X \rightarrow B$ is a zeroth order reaction. If the initial concentration of X is 0.2 M, the half-life is 6 h. When the initial concentration of X is 0.5 M, the time required to reach its final concentration of 0.2 M will be :-

[JEE (Main) 2019]

(1) 18.0 h (2) 7.2 h (3) 9.0 h (4) 12.0 h

CCK040

8. Which of the following statement is not true for radioactive decay ?

[JEE (Main) 2025]

(1) Amount of radioactive substance remained after three half lives is $\frac{1}{8}^{th}$ of original amount.

(2) Decay constant does not depend upon temperature.

(3) Decay constant increases with increase in temperature.

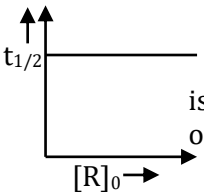
(4) Half life is ln 2 times of $\frac{1}{\text{rate constant}}$.

CCK204

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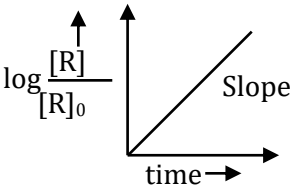
9. Given below are two statements :

Statement (I) :



is valid for first order reaction.

Statement (II) :



is valid for first order reaction

In the light of the above statements, choose the **correct** answer from the options given below :

[JEE (Main) 2025]

(1) Both **Statement I** and **Statement II** are false

(2) **Statement I** is false but **Statement II** is true

(3) Both **Statement I** and **Statement II** are true

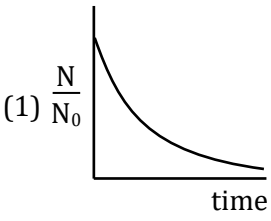
(4) **Statement I** is true but **Statement II** is false

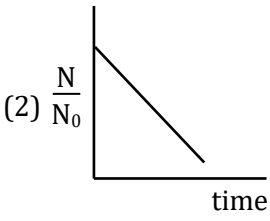
CCK205

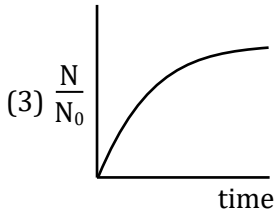
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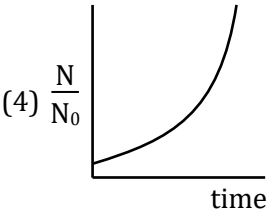
10. For bacterial growth in a cell culture, growth law is very similar to the law of radioactive decay. Which of the following graphs is most suitable to represent bacterial colony growth ?

[JEE (Main) 2025]

(1) 

(2) 

(3) 

(4) 

CCK206

11. The half-life of radioisotopic bromine - 82 is 36 hours. The fraction which remains after one day is _____ $\times 10^{-2}$.

(Given antilog 0.2006 = 1.587) **[JEE (Main) 2024]**

CCK178

12. Time required for completion of 99.9% of a First order reaction is _____ times of half life ($t_{1/2}$) of the reaction. **[JEE (Main) 2024]**

CCK179

13. The rate of first order reaction is 0.04 mol L^{-1} s^{-1} at 10 minutes and 0.03 mol L^{-1} s^{-1} at 20 minutes after initiation. Half life of the reaction is _____ minutes. (Given log2=0.3010, log3=0.4771)

[JEE (Main) 2024]

CCK180

14. $t_{87.5}$ is the time required for the reaction to undergo 87.5% completion and t_{50} is the time required for the reaction to undergo 50% completion. The relation between $t_{87.5}$ and t_{50} for a first order reaction is $t_{87.5} = x \times t_{50}$

The value of x is_____ (Nearest integer) **[JEE (Main) 2023]**

CCK147

15. A and B decompose via first order kinetics with half-lives 54.0 min and 18.0 min respectively. Starting from an equimolar non-reactive mixture of A and B, the time taken for the concentration of A to become 16 times that of B is _____ min. (Round off to the Nearest Integer). **[JEE (Main) 2021]**

CCK048

16. Sucrose hydrolyses in acid solution into glucose and fructose following first order rate law with a half-life of 3.33 h at 25 °C. After 9 h, the fraction of sucrose remaining is *f*. The value of $\log_{10}\left(\frac{1}{f}\right)$ is _____ $\times 10^{-2}$. (Rounded off to the nearest integer) **[JEE (Main) 2021]**

[Assume : ln 10 = 2.303, ln 2 = 0.693]

CCK049

17. A flask contains a mixture of compounds A and B. Both compounds decompose by first-order kinetics. The half-lives for A and B are 300 s and 180 s, respectively. If the concentrations of A and B are equal initially, the time required for the concentration of A to be four times that of B(in s) : (Use ln 2 = 0.693) **[JEE (Main) 2020]**

(1) 180 (2) 120 (3) 300 (4) 900

CCK043

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For More Join: @IITJEE_Advanced

18. For a first order reaction, A → P, $t_{1/2}$ (half life) is 10 days. The time required for $\frac{1}{4}^{th}$ conversion of A (in days) is :- (ln 2 = 0.693, ln 3 = 1.1) **[JEE (Main) 2018]**

(1) 5 (2) 4.1 (3) 3.2 (4) 2.5

CCK038

19. Integrated rate law equation for a first order gas phase reaction is given by (where P_i is initial pressure and P_t is total pressure at time t) **[JEE (Main) 2024]**

(1) $k=\frac{2.303}{t}\times\log\frac{P_i}{(2P_i-P_t)}$ (2) $k=\frac{2.303}{t}\times\log\frac{2P_i}{(2P_i-P_t)}$

(3) $k=\frac{2.303}{t}\times\log\frac{(2P_i-P_t)}{P_i}$ (4) $k=\frac{2.303}{t}\times\frac{P_i}{(2P_i-P_t)}$

CCK181

20.

$[A]_0$ / molL^{-1}	$t_{1/2}$ / min
0.100	200
0.025	100

For a given reaction R → P, $t_{1/2}$ is related to $[A]_0$ as given in table : **[JEE (Main) 2025]**

Given : log 2 = 0.30

Which of the following is *true* ?

A. The order of the reaction is $\frac{1}{2}$.

B. If $[A]_0$ is 1M, then $t_{1/2}$ is $200\sqrt{10}$ min

C. The order of the reaction changes to 1 if the concentration of reactant changes from 0.100 M to 0.500 M.

D. $t_{1/2}$ is 800 min for $[A]_0$ = 1.6 M

Choose the ***correct*** answer from the options given below :

(1) A and C only (2) A and B only (3) A, B and D only (4) C and D only

CCK207

21. Consider the following data for the given reaction

$2HI_{(g)} \rightarrow H_{2(g)} + I_{2(g)}$

	1	2	3
HI (mol L^{-1})	0.005	0.01	0.02
Rate (mol $L^{-1}s-1$)	7.5×10^{-4}	3.0×10^{-3}	1.2×10^{-2}

The order of the reaction is _____.

[JEE (Main) 2024]

CCK182

22. For a chemical reaction A + B → Product, the order is 1 with respect to A and B.

Rate mol L^{-1} s^{-1}	[A] mol L^{-1}	[B] mol L^{-1}
0.10	20	0.5
0.40	x	0.5
0.80	40	y

What is the value of *x* and *y*? **[JEE (Main) 2023]**

(1) 80 and 2 (2) 40 and 4 (3) 160 and 4 (4) 80 and 4

CCK146

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23. The reaction : $2N_2O_5(g) \rightarrow 4NO_2(g) + O_2(g)$, follows first order kinetics. The pressure of a vessel containing only N_2O_5 was found to increase from 50 mm Hg to 87.5 mm Hg in 30 min. The pressure exerted by the gases after 60 min. will be (Assume temperature remains constant)

[JEE (Main) 2015]

(1) 106.25 mm Hg (2) 116.25 mm Hg (3) 125 mm Hg (4) 150 mm Hg

CCK034

24. The reaction $A_2 + B_2 \rightarrow 2$ AB follows the mechanism **[JEE (Main) 2025]**

$A_2 \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} A + A$(fast)

$A + B_2 \xrightarrow{k_2} AB + B$ (slow)

$A + B \rightarrow AB$ (fast)

The overall order of the reaction is :

(1) 1.5 (2) 3 (3) 2.5 (4) 2

CCK208

25. For the reaction,

$2H_2(g) + 2NO(g) \rightarrow N_2(g) + 2H_2O(g)$

the observed rate expression is, rate = $k_f[NO]^2[H_2]$. The rate expression of the reverse reaction is :

[JEE (Main) 2020]

(1) $k_b[N_2][H_2O]^2/[NO]$ (2) $k_b[N_2][H_2O]$

(3) $k_b[N_2][H_2O]^2$ (4) $k_b[N_2][H_2O]^2/[H_2]$

CCK047

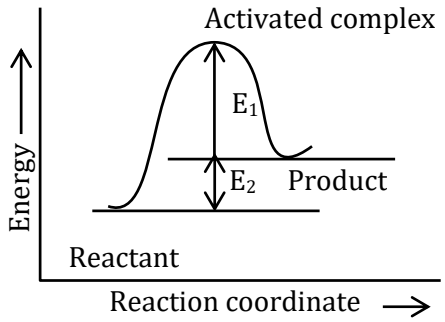
26. $A \rightarrow B$

The molecule A changes into its isomeric form B by following a first order kinetics at a temperature of 1000 K. If the energy barrier with respect to reactant energy for such isomeric transformation is 191.48 kJ mol^{-1} and the frequency factor is 10^{20}, the time required for 50%, molecules of A to become B is _____ picoseconds (nearest integer). **[JEE (Main) 2025]**

[R = 8.314 J K^{-1} mol^{-1}]

CCK209

27. Consider the given figure and choose the **correct** option : **[JEE (Main) 2025]**



(1) Activation energy of backward reaction is E_1 and product is more stable than reactant.

(2) Activation energy of forward reaction is $E_1 + E_2$ and product is more stable than reactant.

(3) Activation energy of forward reaction is $E_1 + E_2$ and product is less stable than reactant.

(4) Activation energy of both forward and backward reaction is $E_1 + E_2$ and reactant is more stable than product.

CCK210

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28. Consider a complex reaction taking place in three steps with rate constants k_1, k_2 and k_3 respectively. The overall rate constant k is given by the expression $k = \sqrt{\frac{k_1 k_3}{k_2}}$. If the activation energies of the three steps are 60, 30 and 10 kJ mol^{-1} respectively, then the overall energy of activation in kJ mol^{-1} is _____. (Nearest integer) **[JEE (Main) 2025]**

CCK211

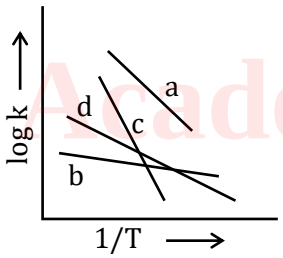
29. For a reaction taking place in three steps at same temperature, overall rate constant $K = \frac{K_1 K_2}{K_3}$. If Ea_1, Ea_2 and Ea_3 are 40, 50 and 60 kJ/mol respectively, the overall Ea is ___ kJ/mol. **[JEE (Main) 2024]**

CCK183

30. If the activation energy of a reaction is 80.9 kJ mol^{-1}, the fraction of molecules at 700 K, having enough energy to react to form products is e^{-x}. The value of *x* is _____.
(Rounded off to the nearest integer) [Use R = 8.31 J K^{-1} mol^{-1}] **[JEE (Main) 2021]**

CCK050

31. Consider the following plots of rate constant versus $\frac{1}{T}$ for four different reactions. Which of the following orders is correct for the activation energies of these reactions? **[JEE (Main) 2020]**

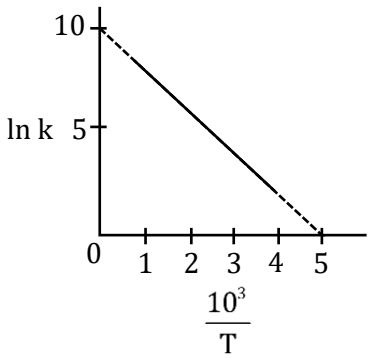


(1) $E_b > E_d > E_c > E_a$ (2) $E_a > E_c > E_d > E_b$

(3) $E_c > E_a > E_d > E_b$ (4) $E_b > E_a > E_d > E_c$

CCK046

32. The rate constant (k) of a reaction is measured at different temperatures (T), and the data are plotted in the given figure. The activation energy of the reaction in kJ mol^{-1} is : (R is gas constant) **[JEE (Main) 2020]**

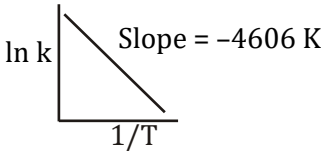


(1) 2R (2) R (3) 1/R (4) 2/R

CCK044

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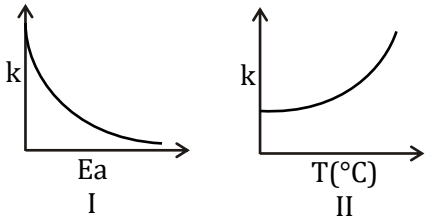
33. For a reaction, consider the plot of ln k versus 1/T given in the figure. If the rate constant of this reaction at 400 K is 10^{-5} s^{-1}, then the rate constant at 500 K is : **[JEE (Main) 2019]**



(1) 2×10^{-4} s^{-1} (2) 10^{-4} s^{-1} (3) 10^{-6} s^{-1} (4) 4×10^{-4} s^{-1}

CCK041

34. Consider the given plots for a reaction obeying Arrhenius equation (0°C < T < 300°C) : (k and E_a are rate constant and activation energy, respectively)



Choose the correct option : **[JEE (Main) 2019]**

(1) Both I and II are wrong (2) I is wrong but II is right

(3) Both I and II are correct (4) I is right but II is wrong

CCK039

35. Two reactions R_1 and R_2 have identical pre-exponential factors. Activation energy of R_1 exceeds that of R_2 by 10 kJ mol^{-1}. If k_1 and k_2 are rate constants for reactions R_1 and R_2 respectively at 300 K, then $\ln(k_2/k_1)$ is equal to :

($R = 8.314$ J $mol^{-1}K^{-1}$) **[JEE (Main) 2017]**

(1) 8 (2) 12 (3) 6 (4) 4

CCK036

36. For the equilibrium, A(g) ⇌ B(g), ΔH is –40 kJ/mol. If the ratio of the activation energies of the forward (E_f) and reverse (E_b) reactions is $\frac{2}{3}$ then :- **[JEE (Main) 2015]**

(1) E_f = 60 kJ/mol; E_b = 100 kJ/mol (2) E_f = 30 kJ/mol; E_b = 70 kJ/mol

(3) E_f = 80 kJ/mol; E_b = 120 kJ/mol (4) E_f = 70kJ/mol; E_b = 30 kJ/mol

CCK035

37. Higher order (>3) reactions are rare due to :- **[JEE (Main) 2015]**

(1) shifting of equilibrium towards reactants due to elastic collision

(2) loss of active species on collision

(3) low probability of simultaneous collision of all the reacting species

(4) increase in entropy and activation energy as more molecules are involved.

CCK033

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EXERCISE - JEE (Advanced) PYQ

1. In dilute aqueous H_2SO_4, the complex diaquodioxalatoferrate (II) is oxidized by $MnO_4^{\ominus}$. For this reaction, the ratio of the rate of change of $[H^{\oplus}]$ to the rate of change of $[MnO_4^{\ominus}]$ is.

[JEE (Advanced) 2015]

CCK075

2. For the elementary reaction M → N, the rate of disappearance of M increases by a factor of 8 upon doubling the concentration of M. The order of the reaction with respect to M is

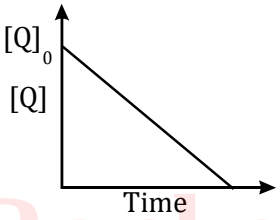
[JEE (Advanced) 2014]

(A) 4 (B) 3 (C) 2 (D) 1

CCK074

3. In the reaction :

$P + Q \longrightarrow R + S$



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the time taken for 75% reaction of P is twice the time taken for 50% reaction of P. The concentration of Q varies with reaction time as shown in the figure. The overall order of the reaction is -

[JEE (Advanced) 2013]

(A) 2 (B) 3 (C) 0 (D) 1

CCK073

4. For the following reaction **[JEE (Advanced) 2021]**

$2X + Y \xrightarrow{k} P$

the rate of reaction is $\frac{d[P]}{dt} = k[X]$. Two moles of X are mixed with one mole of Y to make 1.0 L of solution. At 50 s, 0.5 mole of Y is left in the reaction mixture. The correct statement(s) about the reaction is(are). (Use : ln 2 = 0.693)

(A) The rate constant, k, of the reaction is $13.86 \times 10^{-4}\ s^{-1}$.

(B) Half-life of X is 50s.

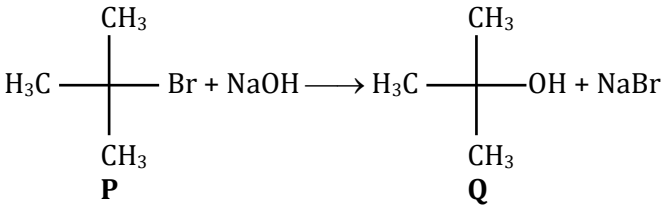
(C) At 50 s, $-\frac{d[X]}{dt} = 13.86 \times 10^{-3}\ mol\ L^{-1}\ s^{-1}$.

(D) At 100 s, $-\frac{d[Y]}{dt} = 3.46 \times 10^{-3}\ mol\ L^{-1}\ s^{-1}$.

CCK083

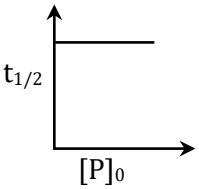
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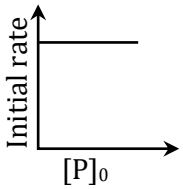
5. Which of the following plots is(are) correct for the given reaction (first order S_N1 reaction)?

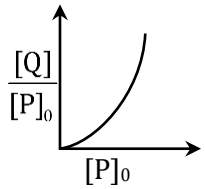


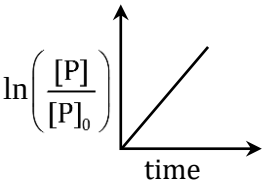
P **Q**

([P]$_0$ is the initial concentration of P) **[JEE (Advanced) 2020]**

(A) 

(B) 

(C) 

(D) 

CCK082

6. The decomposition reaction $2N_2O_5(g) \xrightarrow{\Delta} 2N_2O_4(g) + O_2(g)$ is started in a closed cylinder under isothermal isochoric condition at an initial pressure of 1 atm. After Y × 10^3 s, the pressure inside the cylinder is found to be 1.45 atm. If the rate constant of the reaction is 5 × 10^{-4} s^{-1}, assuming ideal gas behaviour, the value of Y is . **[JEE (Advanced) 2019]**

CCK081

7. For the first order reaction.

$2N_2O_5(g) \longrightarrow 4NO_2(g) + O_2(g)$ **[JEE (Advanced) 2011]**

(A) the concentration of the reactant decreases exponentially with time

(B) the half-life of the reaction decreases with increasing temperature.

(C) the half-life of the reaction depends on the initial concentration of the reactant.

(D) the reaction proceeds to 99.6% completion in eight half-life duration.

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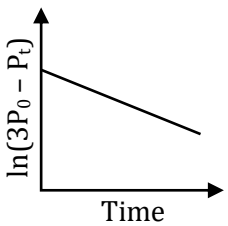
CCK071

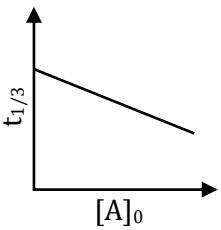
8. An organic compound undergoes first-order decomposition. The time taken for its decomposition to 1/8 and 1/10 of its initial concentration are $t_{1/8}$ and $t_{1/10}$ respectively. What is the value of $\frac{[t_{1/8}]}{t_{1/10}} \times 10$? (take $\log_{10}2 = 0.3$) **[JEE (Advanced) 2011]**

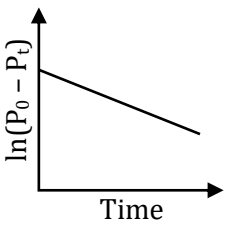
CCK072

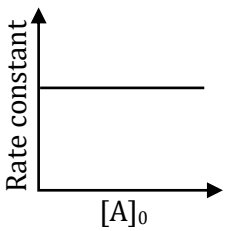
9. For a first order reaction A(g) → 2B(g) + C(g) at constant volume and 300 K, the total pressure at the beginning (t = 0) and at time t are P_0 and P_t, respectively. Initially, only A is present with concentration $[A]_0$, and $t_{1/3}$ is the time required for the partial pressure of A to reach 1/3rd of its initial value. The correct option(s) is (are) :-

(Assume that all these gases behave as ideal gases) **[JEE (Advanced) 2018]**

(A) 

(B) 

(C) 

(D) 

CCK079

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10. Consider the kinetic data given in the following table for the reaction A + B + C → Product.

Experiment No.	[A] (mol dm^{-3})	[B] (mol dm^{-3})	[C] (mol dm^{-3})	Rate of reaction (mol dm^{-3} s^{-1})
1	0.2	0.1	0.1	6.0×10^{-5}
2	0.2	0.2	0.1	6.0×10^{-5}
3	0.2	0.1	0.2	1.2×10^{-4}
4	0.3	0.1	0.1	9.0×10^{-5}

The rate of the reaction for [A] = 0.15 mol dm^{-3}, [B] = 0.25 mol dm^{-3} and [C] = 0.15 mol dm^{-3} is found to be $Y \times 10^{-5}$ mol dm^{-3} s^{-1}. The value of Y is –––––. **[JEE (Advanced) 2019]**

CCK080

11. Consider the following reaction,

$2H_2(g) + 2NO(g) \longrightarrow N_2(g) + 2H_2O(g)$

which follows the mechanism given below:

$2NO(g) \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} N_2O_2(g)$ (fast equilibrium)

$N_2O_2(g) + H_2(g) \xrightarrow{k_2} N_2O(g) + H_2O(g)$ (slow reaction)

$N_2O(g) + H_2(g) \xrightarrow{k_3} N_2(g) + H_2O(g)$ (fast reaction)

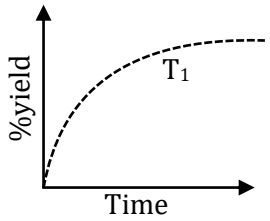
The order of the reaction is_____. **[JEE (Advanced) 2024]**

CCK184

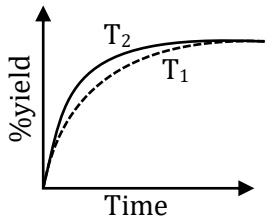
12. The % yield of ammonia as a function of time in the reaction

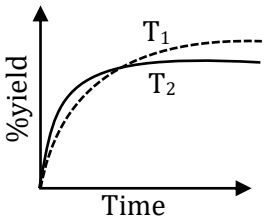
$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g), \Delta H < 0$

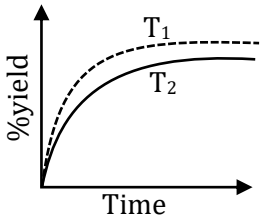
at (P, T_1) is given below -

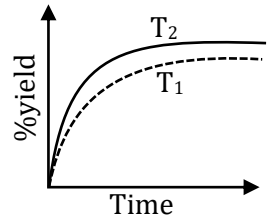


If this reaction is conducted at (P, T_2), with $T_2 > T_1$, the % yield of ammonia as a function of time is represented by - **[JEE (Advanced) 2015]**

(A)


(B)


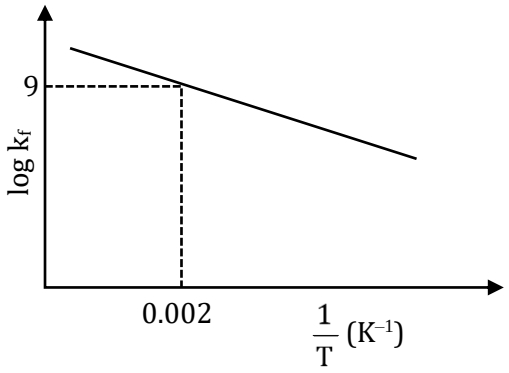
(C)


(D)


CCK076

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13. The plot of log k_f versus $1/T$ for a reversible reaction A (g) $\rightleftharpoons$ P (g) is shown.



Pre-exponential factors for the forward and backward reactions are 10^{15} s^{-1} and 10^{11} s^{-1}, respectively. If the value of log K for the reaction at 500 K is 6, the value of | log k_b | at 250 K is _____.

[K = equilibrium constant of the reaction

k_f = rate constant of forward reaction

k_b = rate constant of backward reaction] **[JEE (Advanced) 2023]**

CCK148

14. In a bimolecular reaction, the steric factor P was experimentally determined to be 4.5. The correct option(s) among the following is(are) : **[JEE (Advanced) 2017]**

(A) The value of frequency factor predicted by Arrhenius equation is higher than that determined experimentally

(B) The activation energy of the reaction is unaffected by the value of the steric factor

(C) Since P = 4.5, the reaction will not proceed unless an effective catalyst is used.

(D) Experimentally determined value of frequency factor is higher than that predicted by Arrhenius equation.

CCK078

15. According to the Arrhenius equation, **[JEE (Advanced) 2016]**

(A) A high activation energy usually implies a fast reaction

(B) Rate constant increase with increase in temperature. This is due to a greater number of collisions whose energy exceeds the activation energy

(C) Higher the magnitude of activation energy, stronger is the temperature dependence of the rate constant

(D) The pre-exponential factor is a measure of the rate at which collisions occur, irrespective of their energy.

CCK077

Parikh's Academy JEE

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ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B	B	C	B	D	C	C	C	A	A
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	D	D	A	C	A	C	D	C	A	C
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	C	B	A	C	A	C	D	C	A	D
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	D	D	C	A	C	B	A	D	A	A
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	D	B	A	B	C	D	D	D	D	A

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	9
Ans.	ABD	ABC	C	AD	AC	C	C	A	A

EXERCISE (S)

Que.	1	2			3	4	5		6	7
Ans.	8	(i) 9	(ii) 36	(iii) 54	12	8	(i) 0	(ii) 5	20	60
Que.	8	9		10	11	12	13	14	15	16
Ans.	5	(i) 36	(ii) 108	10	20	4	3	4	240	3
Que.	17	18	19		20	21	22	23		
Ans.	2	5	(i) 7.20	(ii) 10.00	0.01	0.20	0.01	0.01-0.02		
Que.	24		25							
Ans.	(a) 3.00	(b) 0.75	13.44							

Bankers Academy JEE

EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	2	17	4	3	1	2	1	3	4	4
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	63	10	24	3	108	81	4	2	1	3
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	2	1	1	1	4	69	3	20	30	14
Que.	31	32	33	34	35	36	37			
Ans.	3	1	2	4	4	3	3			

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	8	B	D	BCD	A	2.30	ABD	9	AD	6.75
Que.	11	12	13	14	15					
Ans.	3	B	5	BD	BCD					

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Important Notes

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